

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION with OpenCV

COURSE CODE: DSA0204

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage: 10%

1. An image appears too bright. Explain and justify the most suitable grey level transformation to improve visibility, clearly describing the concept, application process, and expected outcome.
2. A dark image lacks detail. Describe and justify an appropriate grey level transformation to enhance visibility, explaining how it improves low-intensity regions with proper reasoning.
3. An image has poor contrast. Explain a suitable histogram-based technique for contrast enhancement and illustrate how it improves image quality with logical reasoning.
4. A low-contrast image requires uniform intensity distribution. Describe and justify the method you would apply, explaining the concept, working process, and its effect on intensity levels.
5. An image shows uneven brightness across regions. Explain and apply an appropriate enhancement technique to correct non-uniform illumination, providing a structured and accurate explanation.
6. A noisy image contains random intensity variations. Explain and justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction.
7. An image requires noise reduction without significant loss of details. Describe and apply an appropriate filtering approach, explaining how it balances noise removal and detail preservation.
8. An image appears blurred after smoothing. Explain and justify a suitable sharpening method to restore details, including its working principle and expected results.
9. Fine details in an image are not clearly visible. Describe and justify the filtering technique used to enhance details, explaining its effect on high-frequency components.

10. A noisy image must be preprocessed before further analysis. Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification.
11. An image contains high-frequency noise. Explain and justify the appropriate frequency domain filter, including its working and impact on image quality.
12. You need to preserve low-frequency components while removing noise. Describe and apply a suitable filtering technique, explaining how it achieves the desired effect.
13. An image requires edge enhancement in the frequency domain. Explain and justify the filtering approach used, describing its effect on frequency components.
14. A smooth version of an image is required. Describe and justify the frequency domain technique used for smoothing, explaining its operation and results.
15. An object is clearly brighter than the background. Explain and justify the segmentation technique used, including its working and effectiveness.
16. You need to separate foreground and background based on intensity. Describe and apply an appropriate segmentation method, providing a clear explanation of its process.
17. An image requires identification of object edges. Explain and justify a suitable edge detection method, describing its working and effectiveness.
18. Adjacent pixels have similar intensity values. Describe and apply a segmentation technique to group such pixels, explaining the concept and reasoning.
19. Detected edges in an image appear broken. Explain and justify a method to improve edge continuity, including its role in post-processing.
20. You need to extract object outlines from an image. Describe and apply an appropriate technique, explaining how it identifies boundaries effectively.
21. An image suffers from excessive brightness variation between regions. Explain and justify a suitable histogram processing technique to normalize intensity distribution, describing its working and expected outcome.
22. A medical image requires enhancement of subtle intensity differences. Describe and justify a suitable grey level transformation to highlight fine variations, explaining its effect on diagnostic visibility.
23. An image contains salt-and-pepper noise. Explain and justify the most appropriate spatial filter, describing its working mechanism and effectiveness in noise removal.
24. An image appears overly smooth after filtering, losing important features. Explain and justify a sharpening technique to restore details, including its working and impact on edges.

25. An image contains both low-frequency background and high-frequency noise. Describe and justify a filtering approach that preserves important structures while reducing noise, explaining the concept clearly.
26. An image requires enhancement of edges without amplifying noise significantly. Explain and justify a suitable spatial or frequency domain method, describing its operation and expected result.
27. A grayscale image shows poor separation between object and background intensities. Describe and justify a thresholding approach that improves segmentation accuracy, explaining its working.
28. An image contains multiple regions with similar intensity but different textures. Explain and justify a region-based segmentation approach, describing how it groups pixels effectively.
29. Detected edges are noisy and fragmented. Describe and justify a method to refine edges and improve boundary detection, explaining its role in segmentation.
30. An application requires precise object boundary extraction for shape analysis. Explain and justify a suitable segmentation and boundary detection approach, describing its working and effectiveness.

Code of Conduct:

I Bhavya PS(192524017) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Rubrics for Evaluation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning

<i>Understanding of Concepts</i>	<i>25%</i>	<i>Demonstrates thorough, accurate understanding of concepts</i>	<i>Good understanding with minor gaps</i>	<i>Basic understanding with some misconceptions</i>	<i>Poor understanding; major misconceptions</i>
<i>Explanation</i>	<i>25%</i>	<i>Detailed, well-explained answers with relevant examples</i>	<i>Clear explanation with adequate details</i>	<i>Limited explanation; lacks depth</i>	<i>Poor or unclear explanation</i>
<i>Application</i>	<i>20%</i>	<i>Effectively applies concepts with strong analytical ability</i>	<i>Applies concepts with minor errors</i>	<i>Limited application with weak analysis</i>	<i>No application or incorrect analysis</i>
<i>Organization</i>	<i>15%</i>	<i>Well-structured answers with logical flow and coherence</i>	<i>Organized with minor issues in flow</i>	<i>Basic structure, lacks clarity</i>	<i>Poor organization, incoherent answers</i>
<i>Accuracy</i>	<i>10%</i>	<i>Completely accurate and covers all required points</i>	<i>Mostly accurate with minor omissions</i>	<i>Partially correct with missing points</i>	<i>Incorrect answers with major gaps</i>
<i>Clarity</i>	<i>5%</i>	<i>Clear, neat, professional presentation</i>	<i>Understandable with minor issues</i>	<i>Limited clarity, readability issues</i>	<i>Poorly written, difficult to understand</i>

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COURSE NAME: Computer Vision With OpenCV

COURSE CODE: DSAD204

- 1] An image appears too bright explain and justify the most suitable gray level transformation to improve visibility, clearly describing the concept, application, process, and expected outcome.
use Gamma correction ($\gamma > 1$) to reduce brightness it compresses high-intensity pixel values, restores details in bright regions, and improves overall visibility without affecting image quality significantly.
- 2] A dark image lacks detail. Describe and justify an appropriate gray level transformation to enhance visibility explaining how it improves low-intensity regions with proper reasoning.
Apply Gamma Correction ($\gamma < 1$) to brighten low-intensity pixels, it enhances dark regions, improves visibility and reveals hidden details while maintaining a natural appearance.
- 3] An image has poor contrast. explain a suitable histogram based technique for contrast enhancement and illustrate how it improves image quality with logical reasoning.
use Histogram Equalization. it redistributes pixel intensities over the full grayscale range, increasing contrast and making objects easier to distinguish from the background.

- 4] A low-contrast image requires uniform intensity distribution. Describe and justify the method you would apply, explaining the concept, working process, and its effect on intensity levels.

Apply Histogram to spread intensity values uniformly. This improves global contrast and enhances image quality by utilizing the complete intensity range.

- 5] An image requires noise reduction without significant loss of details. Describe and apply an appropriate filtering approach, explaining how it balances noise removal and detail preservation.

Apply a Bilateral filter. It removes noise while preserving edges by considering both pixel intensity differences and spatial distance.

- 6] Fine details in an image are not clearly visible. Describe and justify the filtering technique used to enhance details, explaining its effect on high-frequency components.

Apply a high-pass filter or Laplacian filter. These enhance high-frequency components, making edges and fine image details more visible.

- 7] A noisy image must be preprocessed before further analysis. Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification.

Use a median filter for preprocessing. It effectively removes impulse (salt and pepper) noise while preserving image edges.

- 8] A smooth version of an image is required. Describe and justify the frequency domain technique used for smoothing, explaining its operation and results.
Use High-Pass Filter. It enhances high-frequency components corresponding to edges and fine details making object boundaries more prominent.
- 9] A ~~image~~ requires edge enhancement in the frequency domain. Explain and justify the filtering approach used, describing its effect on frequency components.
Use Gaussian low-pass Filter in the frequency domain. It removes high frequency noise and produces a smoother image.
- 10] An image requires image enhancement of edges without amplifying noise significantly. Explain and justify a suitable spatial or frequency domain method, describing its operation and expected results.
Apply unsharp masking. It sharpens edge while minimizing amplification of background noise.